ARC HOBBY · QUICK REFERENCE

Wood cheat sheet

Hardness: Janka, pounds of force (higher is harder) **Weight**: lb per cubic foot, dried `outdoors` `blotchy stain` `dust mask`

The woods

Pine 380 · 25 lb

Best for: first projects, shelves, painted furniture, practice pieces, jigs.

Avoid: cutting boards, tabletops that get hard wear (it dents), outdoors unless painted.

Cheap, soft, easy to cut and screw. Knots can loosen; seal them before painting.

`blotchy stain`

Poplar 540 · 29 lb

Best for: anything you'll paint, drawer sides, trim, toys.

Avoid: stained show pieces (blotchy, greenish), outdoors.

Smooth, stable, cheap for a hardwood. Takes paint beautifully.

`blotchy stain`

Basswood 410 · 26 lb

Best for: carving and wood burning, laser-cut parts (basswood plywood), models, toys you paint.

Avoid: outdoors, anything that takes hard knocks.

Pale, even and soft, with hardly any grain to fight: the carvers' wood. Basswood plywood is the usual laser sheet.

Red oak 1,290 · 44 lb

Best for: tables, bookcases, frames, stained furniture.

Avoid: outdoors, cutting boards (open pores soak up water and food).

Strong, takes stain evenly, bold grain. Pre-drill every screw.

White oak 1,360 · 47 lb

Best for: outdoor furniture, tabletops, mission style, anything that gets wet.

Avoid: steel screws outdoors (the wood stains black around them): use stainless.

Heavy and rot-resistant. Quarter-sawn boards show silver flecks.

`outdoors`

Hard maple 1,450 · 44 lb

Best for: cutting boards, workbench tops, children's blocks, anything that takes a beating.

Avoid: staining (blotchy), small hand tools only (it's hard work).

Very hard, pale, food-safe. Burns if the saw blade is dull.

`blotchy stain`

Cherry 950 · 35 lb

Best for: fine furniture, boxes, cutting boards, gifts.

Avoid: stain (blotchy: let it darken by itself), outdoors.

Smooth and easy to work; turns deep red-brown in the light over a year.

`blotchy stain`

Walnut 1,010 · 38 lb

Best for: fine furniture, boxes, cutting boards, contrast strips.

Avoid: outdoors; big projects if money's tight.

Rich brown, works and finishes easily. An oil finish makes it glow.

Mahogany 800 · 37 lb

Best for: outdoor furniture, boats, fine furniture, carving.

Avoid: wood without a sustainable-forest label (FSC): ask.

Stable, reddish brown, holds up outdoors. Interlocked grain can tear out: use sharp tools.

`outdoors`

Western red cedar 350 · 23 lb

Best for: birdhouses, planters, fences, outdoor benches, closet linings.

Avoid: cutting boards and toys, anything that gets knocked about (it dents).

Light, smells great, resists rot and bugs. Use stainless or coated screws.

`outdoors` `dust mask`

Plywood 660 · 34 lb

Best for: shop furniture, jigs, cabinet backs, painted projects.

Avoid: outdoors (unless exterior grade), showing raw edges (band them).

Flat and cheap. The face layer is thin: sand lightly.

`blotchy stain`

Birch plywood ≈1,260 · 42 lb

Best for: cabinets, drawers, jigs, shelves, laser-cut parts (thin sheets).

Avoid: outdoors (unless exterior grade), showing raw edges (band them).

Flat, strong, doesn't move with the seasons. The edge shows its layers.

`blotchy stain`

MDF — · 47 lb

Best for: painted cabinet parts, jigs, speaker boxes, templates.

Avoid: anything that gets wet, screws near an edge (it splits), sagging shelves.

Dead flat and cheap, heavy, paints smooth. No grain.

`dust mask`

Particleboard — · 45 lb

Best for: cabinet boxes and shelves under laminate or veneer, painted shop furniture.

Avoid: anything wet, screws near an edge, long shelves (they sag), screws taken out and put back in.

Cheap and flat, but weak at its edges: confirmat screws and cam locks hold it best.

`dust mask`

Zebrawood 1,830 · 50 lb

Best for: accents: box lids, inlays, pen blanks, a contrast strip.

Avoid: whole projects (costly), food, beginners' hand tools.

Bold stripes, hard, can tear out. Dust irritates eyes and skin.

`dust mask`

Padauk 1,970 · 47 lb

Best for: bright accents, contrast strips, small boxes.

Avoid: outdoors (the colour fades to brown), food.

Orange-red, oily: wipe joints with acetone before gluing. Its red dust stains.

`dust mask`

Board sizes: name and real size

NAME	REAL SIZE	COMMON USE
1×2	¾″ × 1½″	cleats, trim, frames
1×3	¾″ × 2½″	face frames, slats
1×4	¾″ × 3½″	aprons, rails, small shelves
1×6	¾″ × 5½″	birdhouses, box sides, shelves
1×8	¾″ × 7¼″	bookshelf shelves
1×10	¾″ × 9¼″	deep shelves, bench seats
1×12	¾″ × 11¼″	bookcase sides, wide shelves
2×2	1½″ × 1½″	table legs, frames
2×4	1½″ × 3½″	workbenches, frames, studs
2×6	1½″ × 5½″	bench tops, outdoor furniture
4×4	3½″ × 3½″	posts, thick legs
¼×4	¼″ × 3½″	box bottoms, trays
½×4	½″ × 3½″	drawer and box sides

Sold in 6, 8, 10 and 12 ft lengths (2× also 16 ft). The name is the size before drying and planing; the real size is what you measure.

Hardwood at a lumberyard

THICKNESS	ROUGH	PLANED (S2S)
4/4 ("four quarter")	1″	about 13⁄16″
5/4	1¼″	about 11⁄16″
6/4	1½″	about 15⁄16″
8/4	2″	about 1¾″

Random widths and lengths, priced by the board foot. Home stores also sell hardwood planed to ¾″ in 1×4, 1×6 and so on, like pine.

Plywood and sheets

NAME	REALLY	COMMON USE
¼″ plywood	about 7⁄32″ (5.5 mm)	drawer bottoms, cabinet backs
½″ plywood	about 15⁄32″ (12 mm)	drawer sides, small boxes
¾″ plywood	about 23⁄32″ (18 mm)	cabinets, shelves, desks
Baltic birch	6, 12, 18 mm	jigs, drawers, laser parts
⅛″ plywood	about 3 mm	laser parts, box bottoms
¼″, ½″, ¾″ MDF	as named	painted parts, jigs
⅛″ hardboard, ¼″ pegboard	⅛″, about 3⁄16″	backs, drawer bottoms, tool walls: brads and glue
7⁄16″, ¾″ OSB	7⁄16″, 23⁄32″	shop floors, sheds, rough shelves
¾″ melamine	¾″	closet shelves, cabinet boxes: screw it, don't glue its faces; band the edges
¾″ blockboard	¾″	long shelves and doors that stay flat

Full sheets are 4 × 8 ft; stores also sell 4 × 4 and 2 × 4 ft project panels. Baltic birch often comes 5 × 5 ft. A dado for "¾"" plywood must be cut to its real thickness.

Board feet

One board foot is a piece 12″ × 12″ × 1″ (rough thickness).

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thick (in) × wide (in) × long (in) ÷ 144
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A 4/4 board 6″ wide and 8 ft long: 1 × 6 × 96 ÷ 144 = **4 board feet**.

Buy about 25% more than your parts add up to: random widths and planing waste some.

Prices change by store and season: Arc's shopping list starts with typical ones, so type your store's over them. Arc never looks them up.

Pilot holes for #8 screws

Softwoods (pine, cedar) and poplar: **3⁄32″**. Hardwoods and plywood: **⅛″**. Clearance through the first board: **11⁄64″**. Very hard woods (hard maple, zebrawood and padauk): drill slower and always pre-drill.

Outdoors

White oak, mahogany and western red cedar last outside; everything else needs paint or an exterior finish and will still wear. Use stainless or coated screws outdoors.

Stain

Pine, poplar, hard maple, cherry, birch plywood and blockboard go blotchy: use a pre-stain conditioner or a gel stain, and try it on an offcut first. Oak and walnut take stain evenly.

Dust

Wear a dust mask for sanding, and for cutting western red cedar, mdf, hardboard, osb, melamine, particleboard, zebrawood and padauk: their dust can irritate eyes, skin and lungs.

Hardness and weight are typical published values; real boards vary. Check anything that matters with your store or a wood teacher. The pictures are drawn by code: colour and figure change from board to board.